ADS2001

Virtual Internship

Predicting final scores based on team discourse

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Table of Contents

1. Background
2. Problem Statement
3. Data Overview
4. Data Cleaning
5. Data Engineering
6. Exploratory Data Analysis
7. Predictive Modelling
8. Conclusion

01. Background

Background

Nephortex Virtual Internship

1. First year students design nanotech membranes for dialysis

2. Understand Engineering Tasks & Career Persistence

BME
BIOMEDICAL ENGINEERING

Teams Allocation

STUDENTS

3- 7 per team

MENTORS

1 - 2 per team

AIM

1. Analyze **team chat data**

2. Find **communication factors** influencing the final report scores

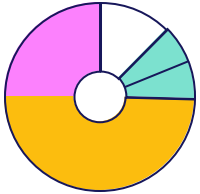
02. Problem Statement

Problem statement

1.

Communication Pattern and Student Performance

How communication pattern within student teams affect their performance in the final design report?



2.

Mentor effect

How does mentor involvement influence the final outcome score for student reports?

03. Data Overview

Data overview

`virtualInternshipData_ADS2001.csv`

	userIDs	implementation	Line_ID	ChatGroup	content	group_id	RoleName	roomName	m_experimental_testing	m_making_design_choices	m_asking_questions	j_customer_consultants_requests	j_performance_parameters_requirements	j_communication	OutcomeScore	wordCount
0	1	a	1	PRNLT	Hello team. Welcome to Nephrotex!	2	Mentor	Introduction and Workflow Tutorial with Entran...	0	0	0	0	0	0	4	5
1	1	a	2	PRNLT	I'm Maria Williams. I'll be your design adviso...	2	Mentor	Introduction and Workflow Tutorial with Entran...	0	0	0	0	0	0	4	11
2	1	a	3	PRNLT	I'm here to help if you have any questions.	2	Mentor	Introduction and Workflow Tutorial with Entran...	0	0	0	0	0	0	4	9

- Chat history from 75 student teams collected across 15 implementations in Nephrotex - to be converted into team-level later
- Each entry corresponds to a **single message** and **its characteristics** - Most variables are categorical
- **16** variables, **19180** observations

04. Data Cleaning

Data cleaning

- Data quality: The data is mostly clean and has no formatting errors
- Only **3 missing values** were found in `RoleName` from the *same user*

By looking at a message by the user, *"I completed the interview but I can't find where the notebook is..."*

Indicates a **Player** rather than a Mentor

- So the `RoleName` is manually imputed as "Player"

	userIDs	implementation	Line_ID	ChatGroup	content	group_id	RoleName	roomName	m_experimental_testing	m_making_design_choices	m_asking_question
7765	158	f	7769	PSF	Checking in	4	NaN	Introduction and Workflow Tutorial with Entran...	0	0	
7771	158	f	7775	PSF	Hey, I'm Rylee	4	NaN	Introduction and Workflow Tutorial with Entran...	0	0	
7801	158	f	7805	PSF	I completed the interview but I can't find whe...	4	NaN	Graphing Surfactant Data	0	0	

	userIDs	implementation	Line_ID	ChatGroup	content	group_id	RoleName	roomName	m_experimental_testing	m_making_design_choices	m_asking_qu
7765	158	f	7769	PSF	Checking in	4	Player	Introduction and Workflow Tutorial with Entran...	0	0	
7771	158	f	7775	PSF	Hey, I'm Rylee	4	Player	Introduction and Workflow Tutorial with Entran...	0	0	
7801	158	f	7805	PSF	I completed the interview but I can't find whe...	4	Player	Graphing Surfactant Data	0	0	

05. Data Engineering

Data Engineering

Before Data Engineering

Features Engineering!

	Model	Train Accuracy	Test Accuracy	Train Macro F1	Test Macro F1
0	Logistic Regression	0.467	0.533	0.402	0.348
1	Decision Tree	1.000	0.333	1.000	0.345
2	KNN	0.500	0.267	0.314	0.173
3	Random Forest	1.000	0.267	1.000	0.149

Data Engineering

Word Count

- Identify the relationship between length of text and final score

1 2 3 4 5 6 7 8 9 10

5

1. char_count

Counts the number of characters in a sentence

2. avg_word_length

Counts the number of words in a sentence then average out.

3. message_length_category

Short	Medium	Long	
≤20	21- 49	≥50	avg_word_length

Data Engineering

Natural Language Processing (NLP)

- Enables computers to understand, interpret, and generate human language
- Apply NLP on 'content' feature to create more feature

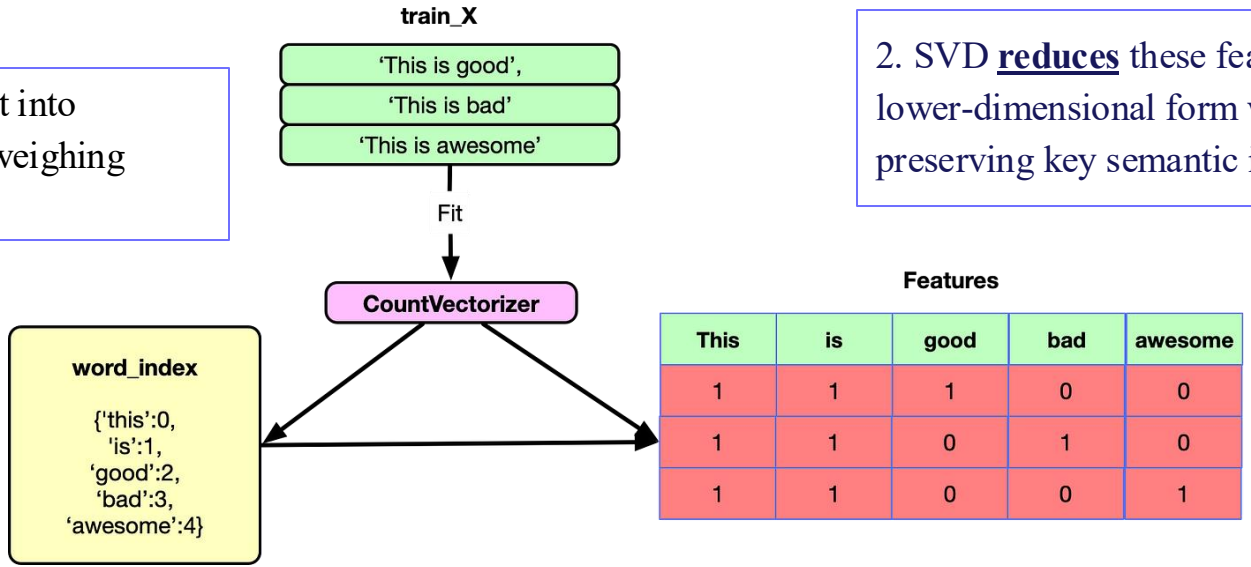
ChatGroup	content	group_id	RoleName
PRNLT	Hello team. Welcome to Nephrotex!	2	Mentor
PRNLT	I'm Maria Williams. I'll be your design adviso...	2	Mentor
PRNLT	I'm here to help if you have any questions.	2	Mentor

Data Engineering

TF-IDF Vectorization and Singular Value Decomposition (SVD)

1. TF-IDF converts text into numerical features by weighing word importance

2. SVD **reduces** these features into a lower-dimensional form while preserving key semantic information



This	is	good	bad	awesome
1	1	1	0	0
1	1	0	1	0
1	1	0	0	1

Data Engineering

Lexical Diversity

Measures richness of vocabulary (unique words) in the chat

0

1

Close to 0, low diversity

Close to 1, high diversity

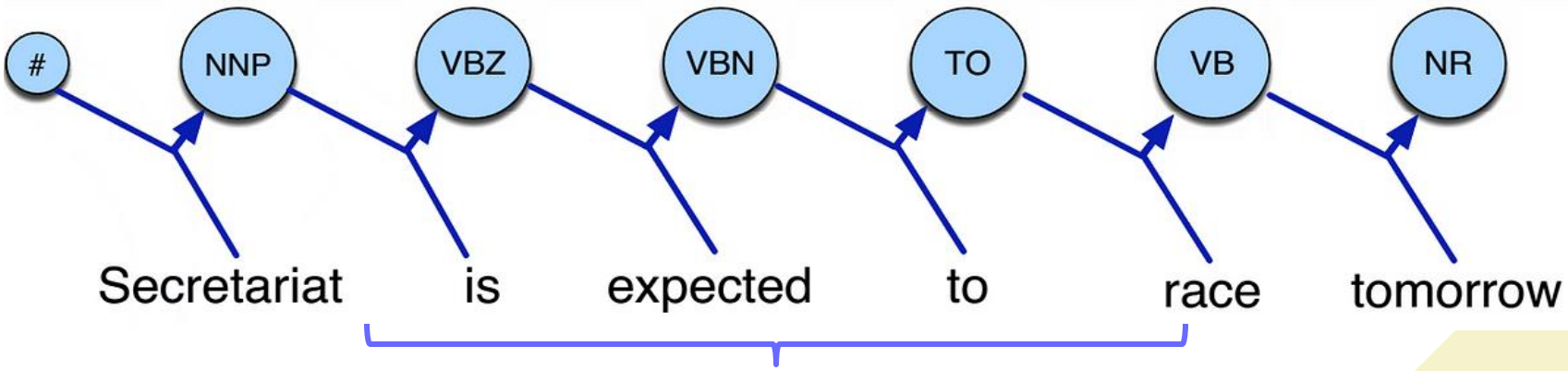
Data Engineering

Topic Coherence

Uses two metrics to calculate a score

1. Entropy — how diverse the topic spread is

2. Cosine Similarity — how similar the discussion topics are



Randomness

Data Engineering

Sentiment Analysis

Detects emotions or opinions in text

Applied to chat logs between players and mentors

Positive

Neutral

Negative

One-hot encoded

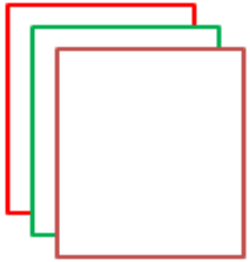
sentiment_consistency

uses standard deviation to measure consistency of sentiment

Data Engineering

Latent Dirichlet Allocation (LDA)

Used to uncover topics in chat messages



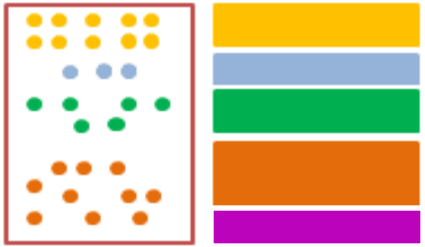
Collection of Text Documents

1. Text is transformed into a Bag-of-Words matrix using CountVectorizer.

Topic Modeling

Cluster of word by topic

2. Engineered topics by clustering



Cluster of document by topic

5 topics

- Topic 0
- Topic 1
- Topic 2
- Topic 3
- Topic 4

3. Topic features are added back to the dataset to reveal thematic patterns in the conversations.

Team level statistics

	num_players	implementation	ChatGroup	m_experimental_testing	m_making_design_choices	m_asking_questions	j_customer_consultants_requests	j_performance_parameters_requirements	j_communication	wordCou
0	7	a	PRNLT	11.0	30.0	64.0	3.0	23.0	7.0	4283
1	6	a	PMMA	7.0	25.0	35.0	4.0	13.0	8.0	3750
2	7	a	PSF	26.0	33.0	69.0	3.0	20.0	10.0	4441

sentiment_negative	sentiment_neutral	sentiment_positive	message_length_category_long	message_length_category_medium	message_length_category_short	avg_outcome_score	mentor_player_chat_ratio	topic_0	topic_1	topic_2	topic_3	topic_4
39.0	161.0	145.0	9.0	47.0	289.0	4.0	0.42	0.245215	0.437013	0.260773	0.236215	0.178548
10.0	101.0	105.0	10.0	60.0	146.0	4.0	0.53	0.256311	0.403897	0.217011	0.261177	0.164208
40.0	153.0	126.0	8.0	55.0	256.0	3.0	0.37	0.248762	0.440556	0.200882	0.287243	0.181767

topic_coherence	sentiment_consistency	team_lexical_diversity	team_char_count	team_word_count	team_avg_word_length
0.162595	0.161869	0.944966	67.733333	12.426087	4.358511
0.168917	0.188687	0.955132	98.157407	17.361111	4.353220
0.181099	0.149008	0.967919	75.304075	13.921630	4.138796

Aggregated each feature at team-level for each team and added each observation to a newly created team-level data frame.

29 variables

75 observations

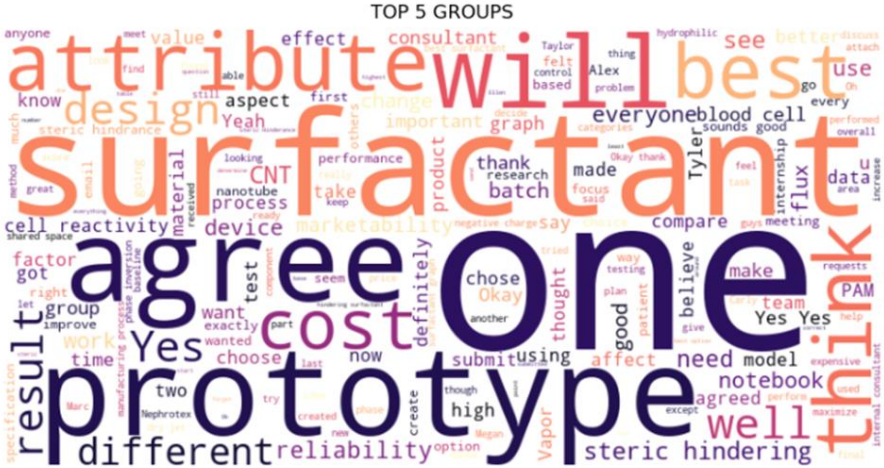
Our target variable:

avg_outcome_score

(ranging from integers **2** to **6**)

06. Exploratory Data Analysis

Word cloud


TOP 5 GROUPS
attribute
will
best
surfactant
design
one
agree
cost
prototype
think
Yes
well
different
reliability
steric hindering
everyone
blood cell
batch
device
compare
need
model
notebook
make
PAM
chose
Okay
thought
believe
good
result
factor
group
time
choose
process
aspect
Yeah
graph
CNT
nanotube
cell reactivity
material
product
flux
internship
data
Tyler
Yes Yes
team
meeting
marketability
affect
submit
using
high
Vapor
agreed
consultant
effect
see
use
know
important
performance
thank
research
made
take
test
want
exactly
work
time
two
Nephrotex
create
definitely
seem
patient
give
help
requests
expensive
internal consultant
manufacturing process
specification
hydrophilic
steric hindering
anyone
value
meet
find
Taylor
felt
control
based
Alex
problem
thing
go
every
sounds good
overall
performed
increase
area
keep
said
focus
negative charge
shared space
method
great
email
received
component
phase inversion
limitation
Mary
Megan
new
created
though
try
option
maximize
perform
except
used
internal consultant
manufacturing process
hydrophilic surfactant
specification
steric hindering
flux rate
testing
making
darren
still
highest
slides
probably
made
know
write
summary
Alex
guess
point
question
decide
look
choose
two
low
device
make sure
graph
internship
Hello
make
different
guys
submitted
sure
membrane
material
Let
Agreed
go
Deliverable
biological
work
report
anyone
working
requirements
sounds good
choose
believe
able
stephen
idea
attribute
hydrophilic
cheapest
higher
another
important
alright
cell reactivity
team
shared space
now
want
either
used
right
ok
find
u
problem
high
manufacturing process
right corner
first
Dry Jet
Nephrotex
see
guys
submitted
think
Yes
notebook
cost
need
best
good
reliability
submit
internship
Hello
make
different
see
Nephrotex
guys
submitted
think
Yes
ok
right
used
either
now
want
shared space
team
cell reactivity
work
Deliverable
biological

LDA Model

Top 5 performing teams

Topic 0: attribut | surfact | best | devic | consult | hinder | tri | steric | choos | design
Topic 1: agre | prototyp | steric | think | cnt | best | like | reliabl | cost | use
Topic 2: ye | good | notebook | submit | everyon | batch | sound | need | think | prototyp
Topic 3: surfact | use | categori | graph | best | differ | rate | design | aspect | perform
Topic 4: thank | team | okay | time | work | meet | help | group | new | design

1. Frequent use of technical terms
2. Collaboration present
3. Technical decision-making language

Bottom 5 performing teams

Topic 0: design | team | batch | work | new | need | submit | attach | specif | make
Topic 1: thank | share | space | make | need | okay | let | consult | everyon | meet
Topic 2: prototyp | agre | notebook | devic | hello | use | surfact | good | yeah | tri
Topic 3: surfact | ye | cost | best | cnt | reliabl | good | flux | reactiv | neg
Topic 4: ok | deliver | right | submit | list | notebook | alex | click | internship | guy

1. Casual phrasing
2. Less evaluative language
3. Lack of decision-making

Correlation Analysis

6 discourse features with *some correlation* to the **target** (`avg_outcome_score`)

Use of experimental techniques

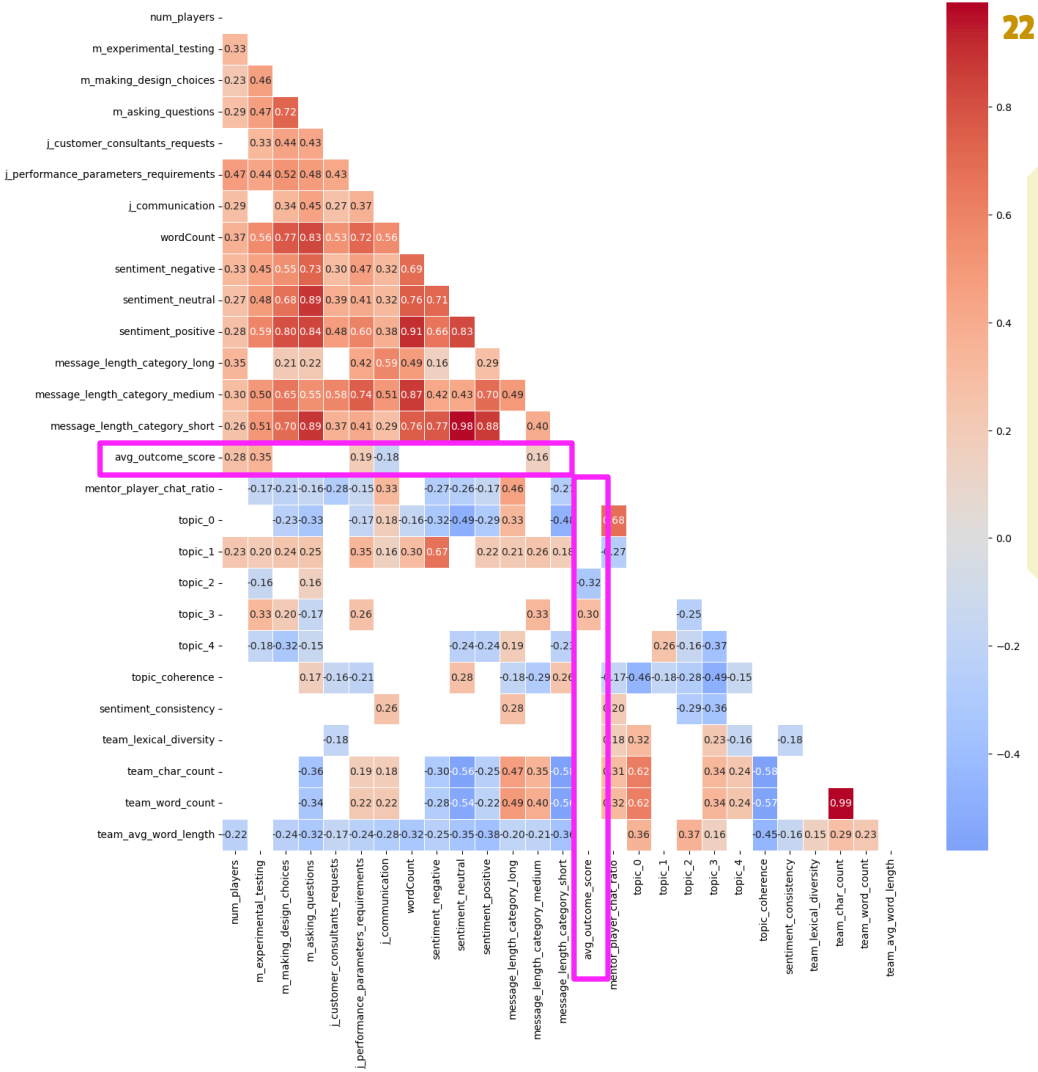
Refer to performance parameters

Medium-length messages

topic_3
(experiments & performance)

topic_2
(agreeable/non-design related)

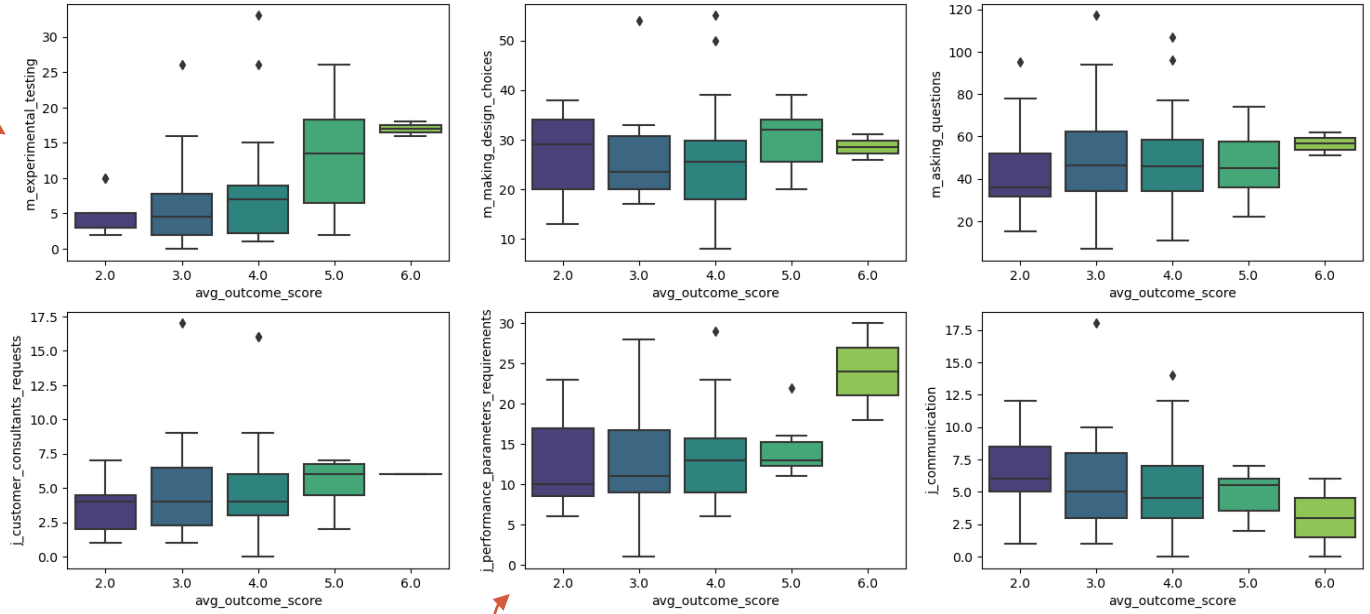
Facilitating communication



Distribution of target across original discourse features

Correlation score: **0.35**

The range of different score classes **overlap** a lot!

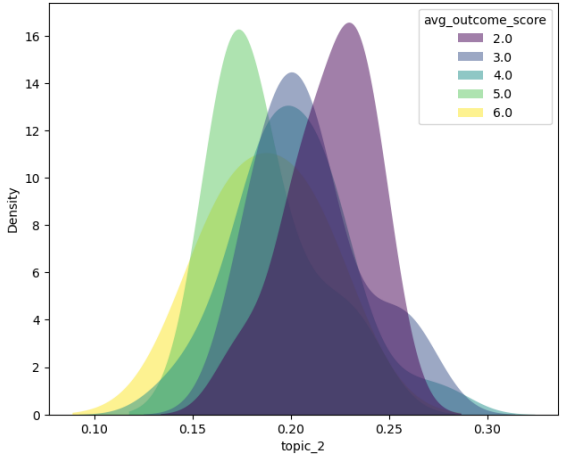


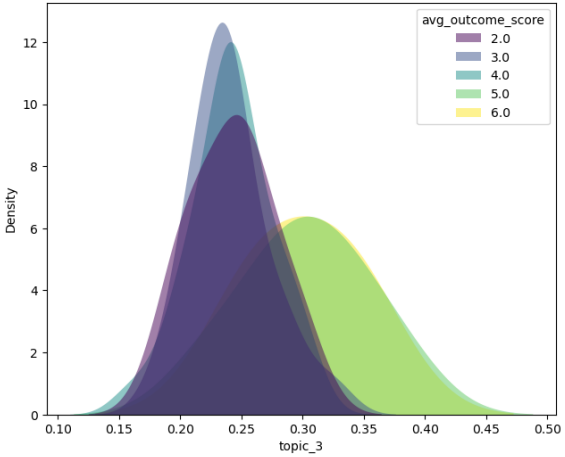
Correlation score: **0.19**

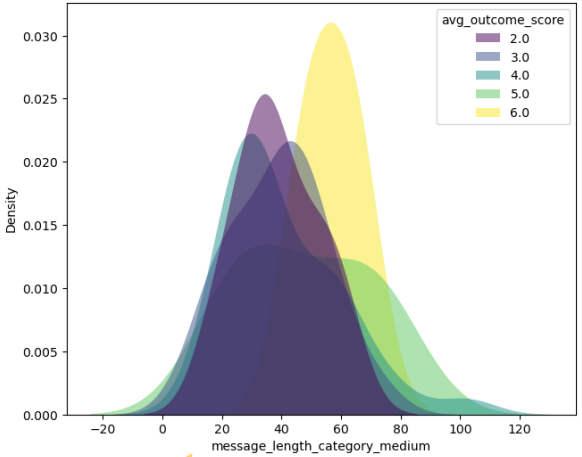
Correlation score: **-0.18**

Distribution of target across correlated engineered features

corr. score: **-0.32** **0.30** **0.16**







Very slight distinction between each score class

Rather than distinction between each class, there is a **distinction between higher score classes** (≥5) vs. **lower score classes** (<5)

07. Predictive Modeling

Baseline model: first attempt of prediction

Problem Statement & Model Type

→ Why classification?

Baseline Models

- Logisitic Regression
- Decision Tree
- KNN
- Random Forest

Summary of Model Performances:

	Model	Train Accuracy	Train Macro F1	Test Accuracy	Test Macro F1
0	Logistic Regression	0.583333	0.440857	0.200000	0.080000
1	Decision Tree	1.000000	1.000000	0.333333	0.146667
2	KNN	0.516667	0.406869	0.133333	0.061538
3	Random Forest	1.000000	1.000000	0.266667	0.088889

- Insufficient Features
- Imbalanced data
- Small Data Size(n=75)

Overfitting

Tried AdaBoost

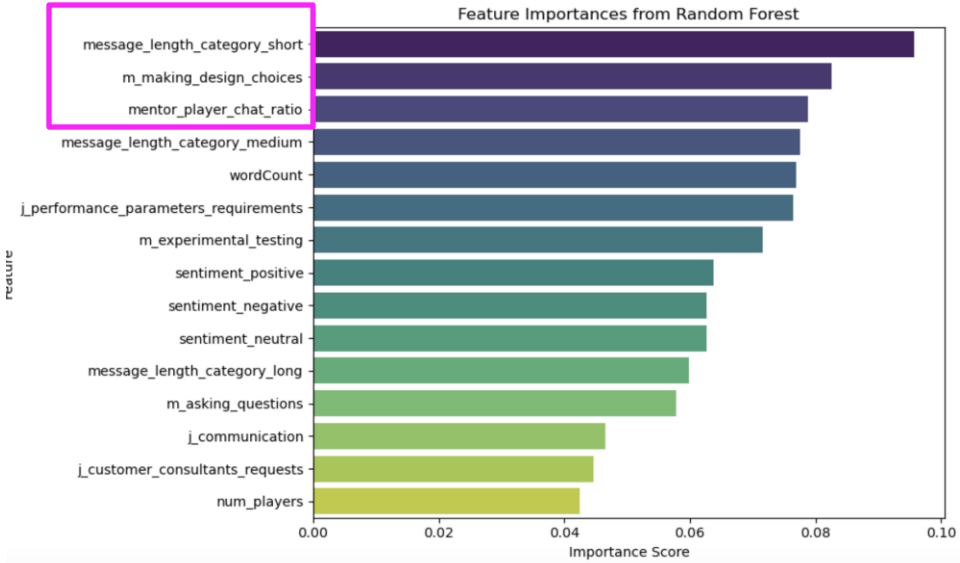
An adaptive ensemble model → Correcting misclassified samples

However...

	accuracy	f1
train	1.000	1.00
test	0.333	0.14

→ Fail to generalize(Overfitting)

Feature Importance →

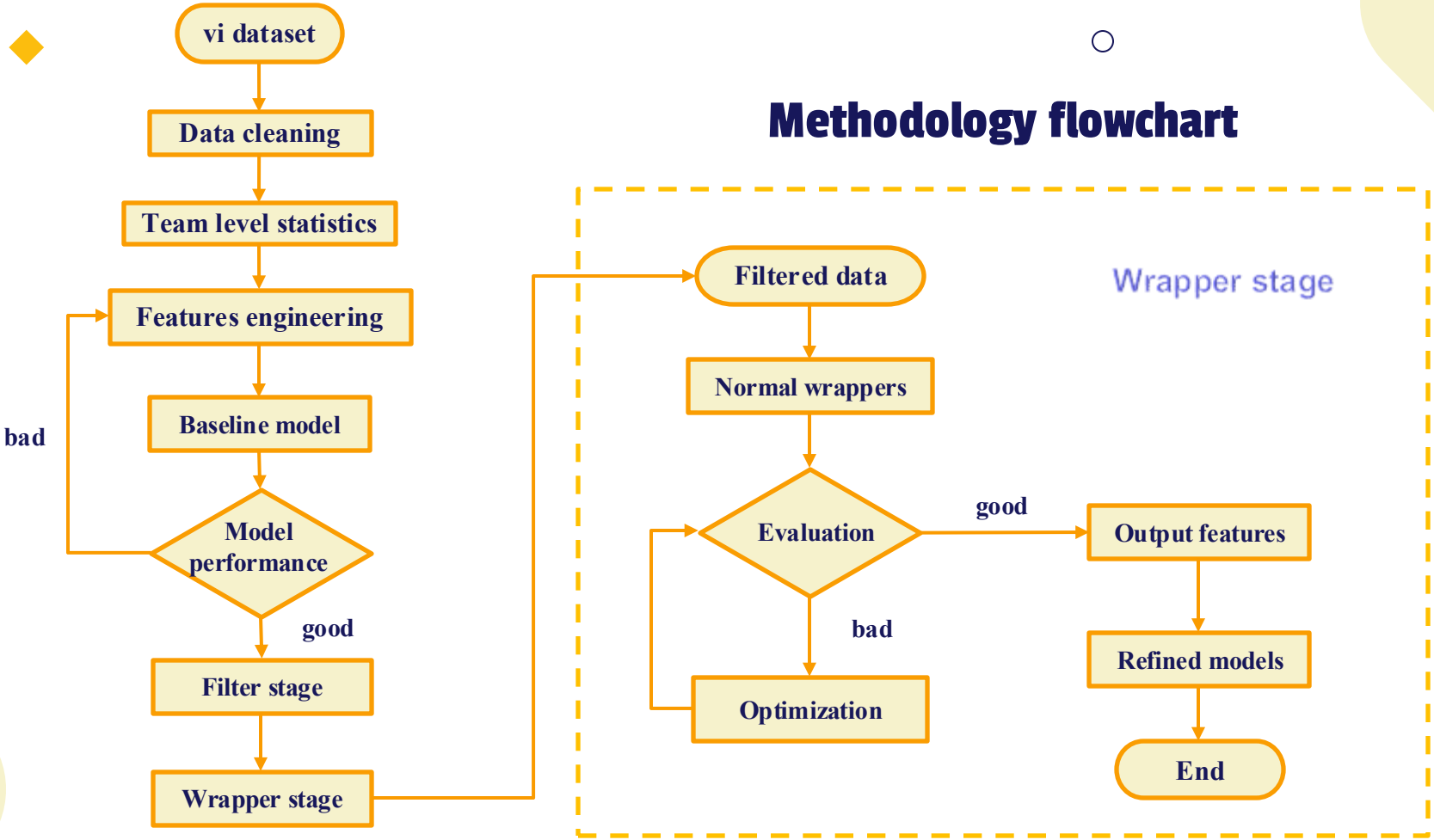


- Maximum Importance score: low (0.094 →≈1.4% more than average)

↓

Need for better feature engineering

Methodology flowchart



Generating more features for Filtering stage

1. TF-IDF Features

Limits top 1000 words as the feature

+1000 features

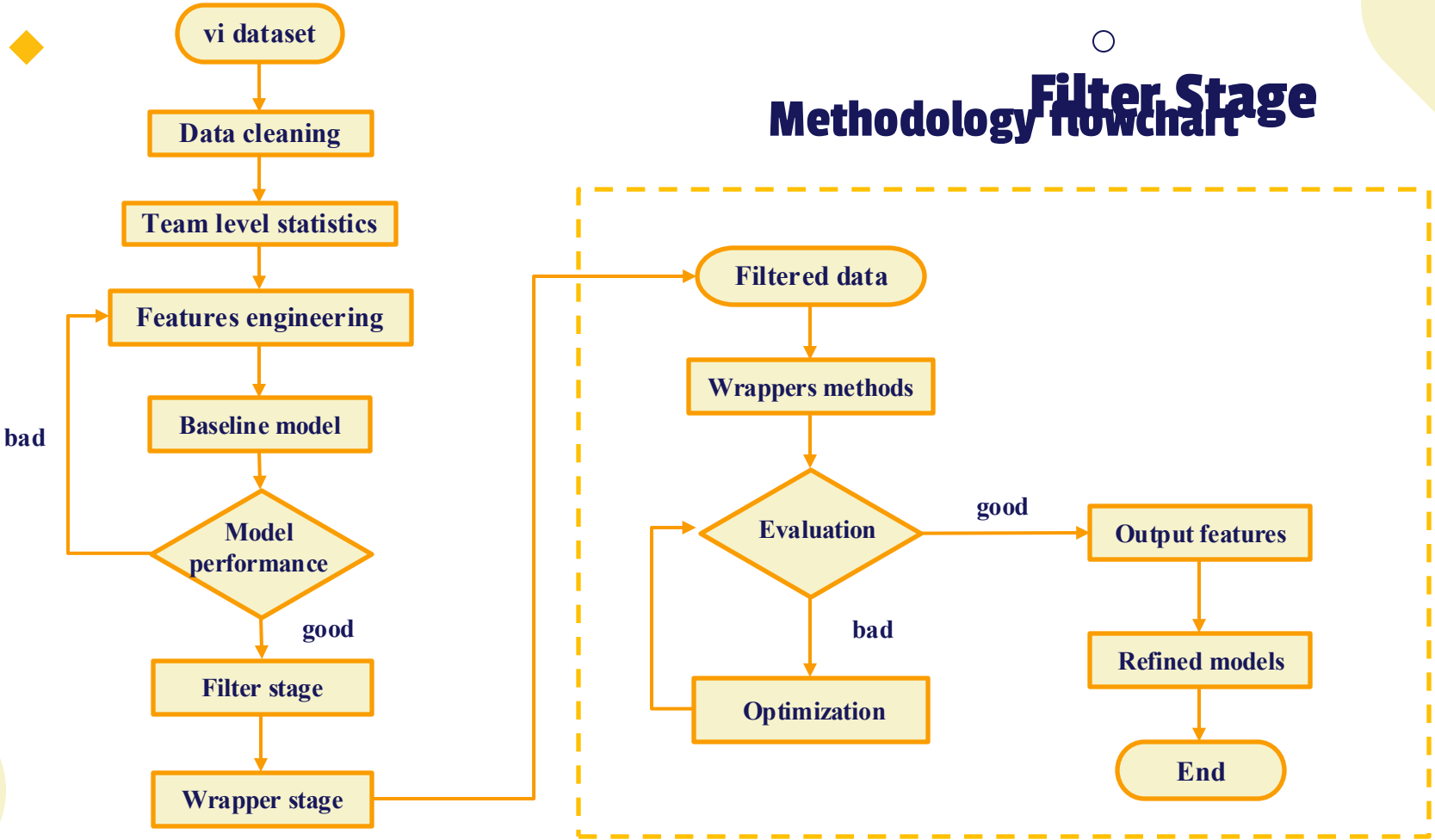
2. Polynomial Interaction Features

Generate pairwise interactions between numeric variables

+231 features

Methodology flowchart

Filter Stage



Filter Stage

Increasing
Testing Score !

0.421

Removed 254 Features

Multicollinearity

Remove High correlated variables

0.474

Removed 25 Features

Variance Threshold

Remove Variables with variance < 0.1

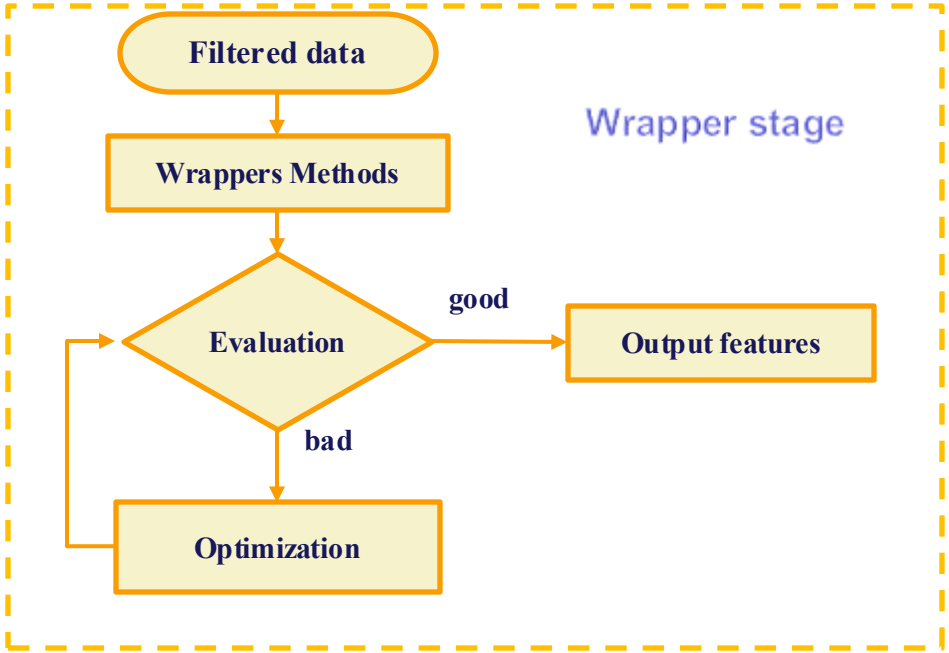
0.526

Removed 537 Features

ReliefF

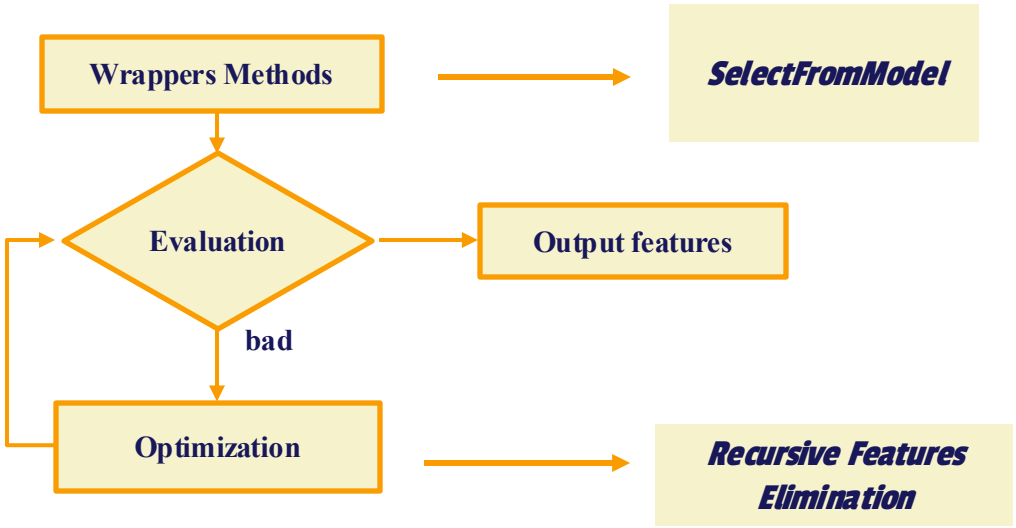
Remove features by comparing features and target with their nearest neighbors

Wrapper Stage



400 features

Wrapper Stage



takes the output of a model and uses attributes of the model to **rank the features** and then extract features based on some **threshold**.

	Accuracy	Precision	Recall
train	1.000	1.000	1.0
test	0.526	0.243	0.3

recursively **eliminates** features until the required number of features

	Accuracy	Precision	Recall
train	0.732	0.491	0.426
test	0.684	0.354	0.400

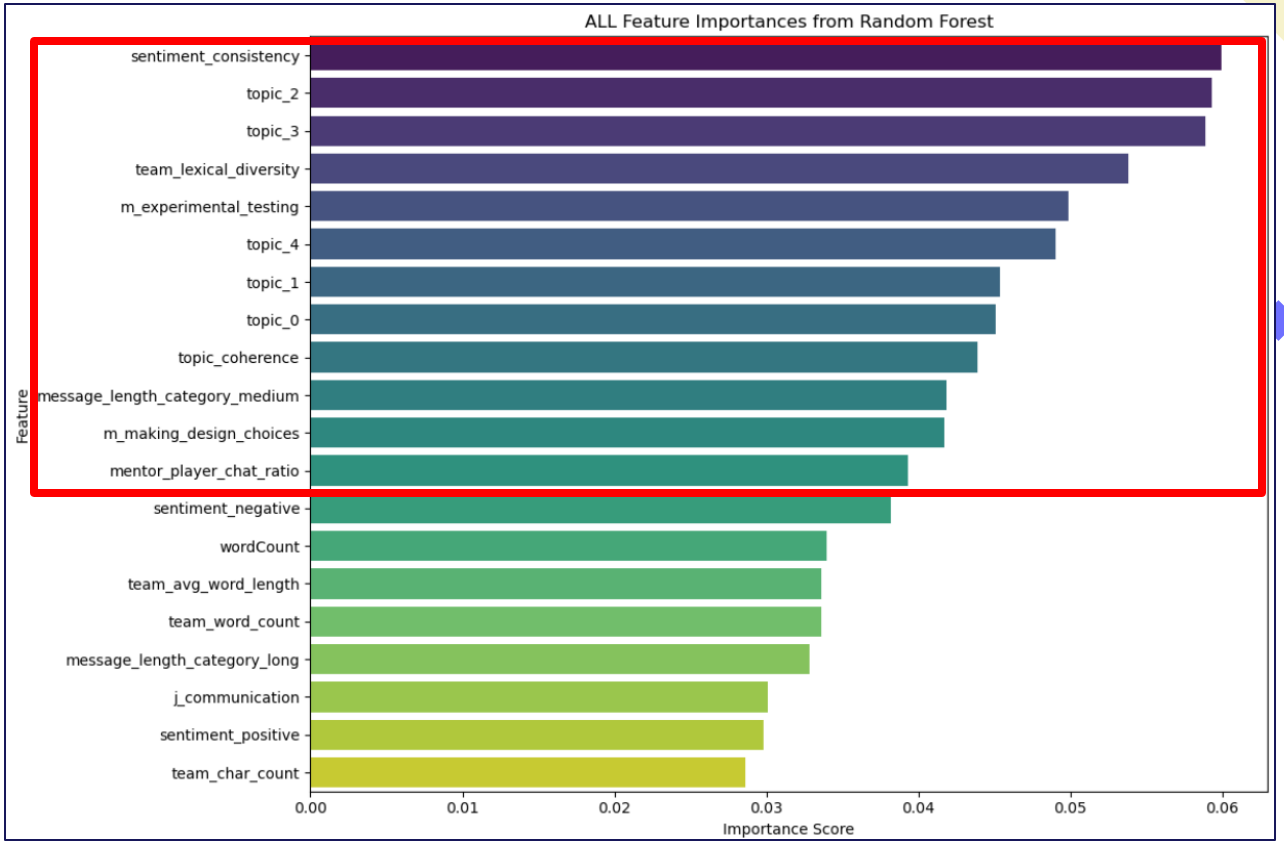
Output Features

Output features

Shared Importance =1/26 ≈ 0.03846

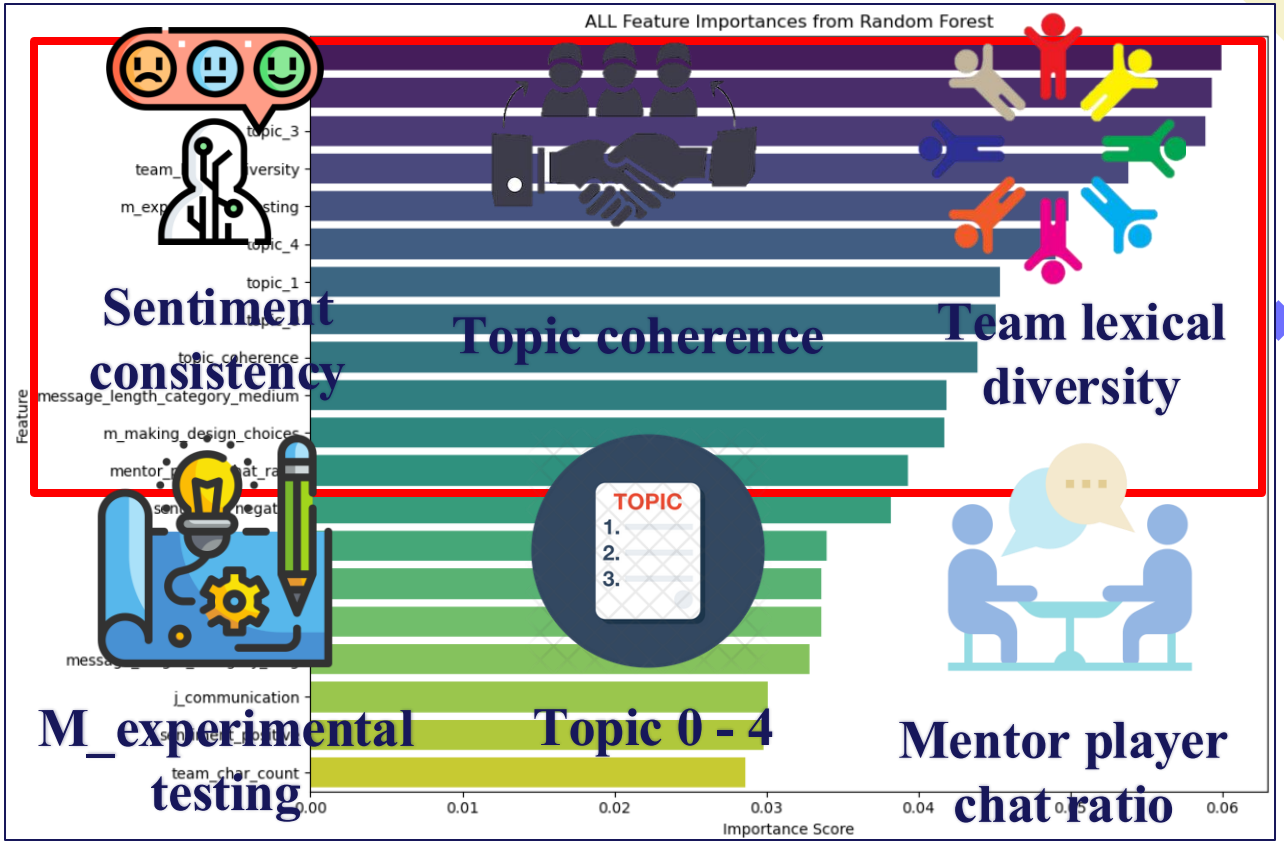
Refined models

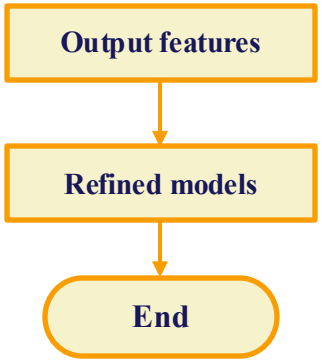
End



Output Features

Shared Importance =1/26 ≈ 0.03846





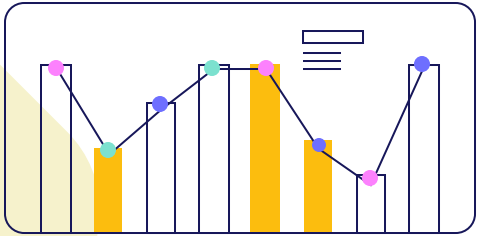
Ada Boost Classifier

- Base model is **Decision Tree Classifier**

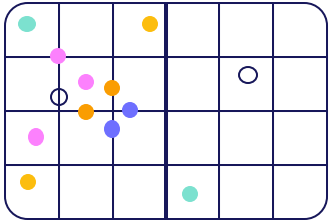
Hyperparameter tuning:

- max_depth = 200
- n_estimator = 200
- min_samples_split = 5

	# accuracy	# f1
train	1.0	1.0
test	0.733	0.282



08. Conclusion



Conclusion

OUR FINDINGS

1. **AdaBoostClassifier**: Best Model After Feature Refinement

2. Key **NLP Features Engineering** Significantly Improve Prediction

3. Collaborative **Communication Pattern** Boosts Performance

4. **Mentor Interaction** has Limited Impact on Performance

Future Suggestions

1. Incorporate **richer NLP Features**

2. Use **network analysis** to identify key contributors

THANK YOU